Virtual Avatar Summary Report

Date: 4th November 2026 **Timestamp:** 18:07:02 PM

Category: Technical

Learner Profile:

Name: Soumiya Ayyappan
Role: Graduate Apprentice | Domain: Graduate Apprentice
Tech Stack:

Initial Q&A Setup:

Q: Which technology or domain do you want to focus on?
A: databricks

Q: Is this for Concept Learning or a live issue?
A: Concept Learning

Session Conversation:

[05:42 PM] Me:
Hello! Here is some background context about me for our session. My primary focus category for this session is technical. I am Soumiya Ayyappan, working as a Graduate Apprentice in the Graduate Apprentice domain. My tech stack includes . Here are some specific details about my project architecture: Regarding "Which technology or domain do you want to focus on?", my answer is "databricks". Regarding "Is this for Concept Learning or a live issue?", my answer is "Concept Learning". Please use this information as context for my questions. Do not use a generic opening greeting. Instead, greet me as Phoebe, briefly summarize that you understand my background and focus area, and ask what we should focus on today!

[05:42 PM] PHOEBE:
Hello Soumiya, I understand you're a Graduate Apprentice focusing on Databricks for concept learning. What specific aspect of Databricks would you like to explore today?

[05:43 PM] Me:
how do you join multiple data free?

[05:43 PM] PHOEBE:
It seems you're curious about the role of a data engineer, along with concepts like DataFrames (DF), Resilient Distributed Datasets (RDDs), and data joining techniques in Databricks. What outcome are you trying to achieve by understanding these concepts?

[05:43 PM] Me:

Learn these concepts.

[05:43 PM] PHOEBE:
Got it! To answer questions about becoming a data engineer, focus on your passion for data, problem-solving skills, and interest in technology.

[05:44 PM] Me:
can we join multiple data frames and how?

[05:44 PM] PHOEBE:
Yes, you can join multiple DataFrames in Databricks using operations like join(). You can specify the join type (inner, outer, etc.) and the condition for joining. How confident do you feel about using these join operations?

[05:44 PM] Me:
Explain catalyst optimization.

[05:44 PM] PHOEBE:
Catalyst optimization is a query optimization framework in Apache Spark that improves the performance of queries by applying various optimization techniques. It analyzes query plans and applies transformations like predicate pushdown and constant folding. Shall we dive deeper into how these transformations work?

[05:44 PM] Me:
Yes.

[05:44 PM] PHOEBE:
Great! Predicate pushdown reduces the amount of data read by filtering rows early in the query execution, while constant folding simplifies expressions by evaluating constant values at compile time. How confident do you feel about applying these concepts in your queries?

[05:45 PM] Me:
You will do?

[05:45 PM] PHOEBE:
Mediterranean architecture refers to a design approach that emphasizes scalability, resilience, and modularity in data processing. In a scenario where a job fails in production, you would first check logs for errors, identify the root cause, and then fix the issue before re-running the job. What specific aspects of handling job failures are you interested in exploring further?

[05:45 PM] Me:
during production, then what you will do?

[05:46 PM] PHOEBE:
In that case, I would first check the monitoring tools for alerts or logs to identify the failure reason. Then, I would assess the impact, notify relevant stakeholders, and initiate a fix or rollback if

necessary. Would you like to explore best practices for monitoring and handling such failures?

[05:46 PM] Me:
What are the tools we used in Goldaire, And What Is Lookup Tables In Goldaire.

[05:46 PM] PHOEBE:
It seems you meant "Goliath" or a similar tool, but I'm not familiar with "Goldaire." Could you clarify what Goldaire refers to? As for lookup tables, they are used to store reference data that can be joined with other datasets to enrich them. How would you like to apply lookup tables in your projects?

[05:46 PM] Me:
null values and the missing values and store in quarantine.

[05:46 PM] Me:
that, Javier, moving on to dashboard. Tell me that one.

[05:47 PM] PHOEBE:
Thank you for clarifying your project context! In a retail sales project using a bronze-silver-gold architecture, the gold layer typically contains cleaned and enriched data ready for analysis. Lookup tables in this context can be used to join additional reference data, like product details or customer information, to enhance insights in your dashboards. What specific insights are you looking to gain from your dashboards?

[05:47 PM] Me:
table, which we have

[05:47 PM] Me:
Cola, and what is the use of that.

[05:48 PM] PHOEBE:
In your scenario, a schema mismatch with product IDs can cause issues when joining tables. Lookup tables help resolve these mismatches by providing a consistent reference for product IDs, ensuring that all related data aligns correctly. They facilitate data integrity and enrich your datasets for accurate analysis. How confident do you feel about implementing lookup tables in your project?

[05:48 PM] Me:
the data tables and create that pipeline?

[05:49 PM] PHOEBE:
Creating a data pipeline from bronze to silver to gold is essential for transforming raw data into a structured format suitable for analysis. Lookup tables enhance this process by providing reference data that can clean and enrich your datasets. They help ensure data consistency and accuracy throughout the pipeline. Would you like to explore how to set up a simple pipeline using these concepts?

[05:49 PM] Me:
Tell me how we are creating the pie.

[05:49 PM] PHOEBE:
To create a data pipeline in Databricks, you typically follow these steps: 1. Extract: Load data from your bronze layer (raw data) into a DataFrame. 2. Transform: Clean and process the data in the silver layer, applying transformations and using lookup tables as needed. 3. Load: Store the processed data in the gold layer for analysis. Would you like to discuss any specific transformation techniques or tools used in this pipeline?

[05:49 PM] Me:
Why is that so? I'm asking.

[05:50 PM] PHOEBE:
Dropping Delta tables may be necessary to avoid data duplication or to refresh data in your pipeline. If a table is no longer needed or has been replaced by an updated version, removing it ensures that your project remains efficient and manageable. Would you like to discuss best practices for managing Delta tables in your pipeline?

[05:50 PM] Me:
Yes.

[05:50 PM] PHOEBE:
Best practices for managing Delta tables include: 1. Version Control: Use Delta Lake's versioning to track changes and rollback if needed. 2. Optimize: Regularly optimize your Delta tables to improve query performance. 3. Vacuum: Periodically run the vacuum command to remove old versions and free up storage. How do you plan to implement these practices in your project?

[05:50 PM] Me:
over here.

[05:51 PM] PHOEBE:
In your project, after processing data through the bronze, silver, and gold layers, you can create dashboards to visualize insights. Lookup tables can be used in the gold layer to enrich your data for better analysis. For forecasting, AI models can analyze historical sales data to predict future trends, helping in inventory management and decision-making. What specific forecasting techniques are you considering for your project?

[05:51 PM] Me:
Forecasting.

[05:52 PM] PHOEBE:
For forecasting, common techniques include time series analysis, regression models, and machine learning algorithms like ARIMA or Prophet. These methods analyze historical data patterns to predict future values. Would you like to explore a specific forecasting method further?

[05:52 PM] Me:
So explain me that what is that procedure.

[05:53 PM] PHOEBE:
In your pipeline, the silver layer serves to clean and refine data from the bronze layer. Quarantine is a temporary storage area where duplicates or problematic records are held for review before being permanently removed. The IKR (Integrity Key Rules) can be used to identify duplicates by checking for unique identifiers or specific data patterns. Once identified, you can decide whether to keep, modify, or discard these records based on your data quality standards. Would you like to discuss how to implement these checks in your pipeline?

[05:53 PM] Me:
stream and then writing that's the right stream data. And then

[05:54 PM] Me:
Tell me wherever the doubts I asked to clear me exactly.

[05:57 PM] PHOEBE:
In the silver layer, you clean and refine the data ingested from the bronze layer. This involves removing duplicates, handling missing values, and storing problematic records in quarantine for further review. You also create transformed tables based on your sales events, inventory movements, and sales returns, ensuring data quality and consistency. In the gold layer, you aggregate and enrich the cleaned data for analysis. This may include applying business logic, joining with lookup tables for additional context (like product details), and preparing data for dashboards or reporting. The gold layer is focused on providing high-quality, ready-to-use datasets for decision-making. Would you like to explore specific techniques for data transformation in the silver layer or aggregation in the gold layer?

[05:57 PM] Me:
elaborate me on that.

[05:59 PM] PHOEBE:
In the silver layer, you're focusing on refining the data from the bronze layer. This includes: 1. Data Cleaning: Removing duplicates and handling missing values. The four notebooks you mentioned likely contain the logic for these processes, ensuring that only high-quality data moves to the next layer. 2. Quarantine Tables: These are used to store records with issues (like duplicates or missing values) for later review. This helps maintain data integrity while allowing you to analyze what went wrong. 3. Final Tables: You create six cleaned tables from the original bronze data, ensuring they are ready for analysis. In the gold layer, the focus shifts to: 1. Aggregation: You perform calculations to summarize the data, such as total sales, average inventory levels, etc. This helps in generating insights. 2. Lookup Tables: These are used to enrich your data with additional context, such as product details. If there's a mismatch in product IDs between tables, it can lead to incomplete joins and inaccurate analysis. 3. Final Output: The gold layer provides a clean, aggregated dataset ready for dashboards and reporting. Understanding the flow from bronze to

silver to gold helps ensure data quality and integrity throughout your project. How do you plan to implement these practices in your current workflow?

[05:59 PM] Me:
better tell.

[06:00 PM] PHOEBE:
Certainly! In the silver layer, you clean the data from the bronze layer by removing duplicates and fixing missing values. This ensures you have high-quality data for analysis. Lookup tables are reference tables that provide additional information, like product details. If there's a mismatch in product IDs between your main tables and the lookup table, it can cause issues when you try to join them, leading to incomplete data in your analysis. In simple terms, if the product ID in your sales data doesn't match the product ID in your lookup table, you won't get the extra details you need for accurate reporting. Would you like to explore how to create and use lookup tables effectively?

[06:00 PM] Me:
Yes.

[06:01 PM] PHOEBE:
To create and use lookup tables effectively, follow these steps: 1. Identify Key Columns: Determine which columns in your main tables (like product IDs) will link to the lookup table. 2. Create the Lookup Table: Compile the reference data (e.g., product details) into a separate table. Ensure it has unique identifiers for accurate joins. 3. Join the Tables: When querying your main data, use a join operation to connect your main table with the lookup table based on the key columns. This enriches your data with additional context. 4. Validate the Join: Check for any mismatches or missing values after the join to ensure data integrity. How do you plan to implement lookup tables in your project?

[06:01 PM] Me:
That's where you do.

[06:02 PM] PHOEBE:
In the gold layer, after cleaning and refining data in the silver layer, you typically perform aggregation and enrichment. This means you summarize the data, such as calculating total sales or average inventory levels. Lookup tables can be used here to add more context, like product descriptions or categories, to your aggregated data. The purpose of the gold layer is to provide a final dataset that is clean, summarized, and enriched, ready for reporting or analysis. So, you first clean and prepare in silver, and then you aggregate and enhance with lookup tables in gold. How do you see this process fitting into your overall project goals?

[06:02 PM] Me:
what I have told, Vincent,

[06:02 PM] Me:
Like, we are doing only aggregation. Why do you need to store that as a table?

[06:03 PM] PHOEBE:
Storing data as fact and dimension tables in a star schema is a common practice in data warehousing. Fact tables contain quantitative data (like sales amounts) and are often aggregated, while dimension tables provide context (like product details or customer information). This structure allows for efficient querying and reporting, as it simplifies complex queries by organizing data logically. By using this approach, you can easily analyze performance metrics and generate insights. Would you like to explore how to design a star schema for your project?

[06:03 PM] Me:
we are doing that in goal

[06:03 PM] Me:
since aggregation is done.

[06:04 PM] PHOEBE:
In the gold layer, you store data as fact and dimension tables to enhance analytical capabilities. Fact tables hold the aggregated metrics (like total sales), while dimension tables provide descriptive context (like product or customer details). This separation allows for more efficient queries and better performance when analyzing large datasets. Even after aggregation, organizing data in this way helps maintain clarity and improves reporting flexibility. How do you plan to utilize these tables in your analysis?

[06:04 PM] Me:
it in a better way.

[06:05 PM] PHOEBE:
Absolutely! The concept of fact and dimension tables is key in data modeling. Fact Tables: These tables store quantitative data for analysis, such as sales amounts or transaction counts. They often contain aggregated metrics and foreign keys that link to dimension tables. Dimension Tables: These tables provide descriptive attributes related to the facts, such as product names, categories, or customer demographics. They help to contextualize the data in the fact tables. This structure allows for efficient data retrieval and analysis, making it easier to generate insights from large datasets. Understanding this will help you navigate your project code effectively. Would you like to explore any specific examples or scenarios related to this concept?

[06:05 PM] Me:
Tell me about that Spark structure streaming

[06:05 PM] Me:
Like, with my dataset?

[06:06 PM] PHOEBE:
Structured Streaming in Spark allows you to process real-time data streams as if they were batch data. In your project, using readStream means you're reading data in real-time, while writeStream allows you to write the processed data to Delta tables. The trigger=“AvailableNow” setting

processes all available data in the stream up to that point and then stops, making it suitable for batch-like processing within a streaming context. In an interview, you could explain that your project uses Structured Streaming to handle real-time data efficiently, allowing you to process batch sales data while leveraging streaming capabilities for timely insights. How do you feel about explaining this concept in your own words?